

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL GCSE

BIOLOGY

Paper 1

Monday 20 May 2024

07:00 GMT

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a ruler with millimetre measurements
- a scientific calculator.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- There are 90 marks available on this paper.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers.

Advice

In all calculations, show clearly how you work out your answer.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	



Answer **all** questions in the spaces provided.

0 1

The sun is the source of light energy for all food chains.

0 1 . 1

Which process do plant cells use light energy for?

[1 mark]

Tick (✓) **one** box.

Decomposition

☐

Photosynthesis

☐

Respiration

☐

0 1 . 2

Which substance in plant cells absorbs light energy?

[1 mark]

Tick (✓) **one** box.

Chlorophyll

☐

Fat

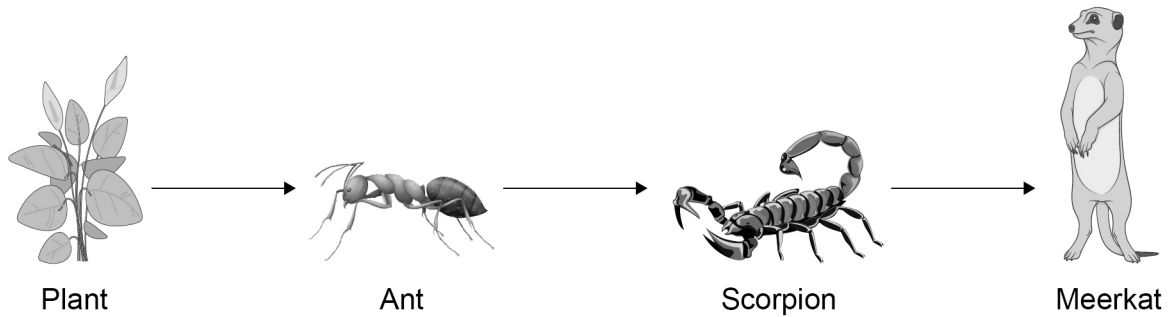
☐

Glucose

☐

Figure 1 shows a food chain.

Figure 1



0 1 . 3 Draw a pyramid of biomass for the food chain in **Figure 1**.

Label the pyramid.

[2 marks]

Question 1 continues on the next page

Turn over ►



0 1 . 4

What is the approximate percentage of biomass transferred between the ant and the scorpion?

[1 mark]

Tick (✓) **one** box.

1%

☐

10%

☐

50%

☐

0 1 . 5

Give **one** way energy is lost between the ant and the scorpion.

[1 mark]



Scorpions can live in deserts.

Figure 2 shows a scorpion in a sandy desert.

Figure 2



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0 1 . 6 Suggest how the claws and the brown body colour help the scorpion survive.

[2 marks]

Claws _____

Brown body colour _____

0 1 . 7 What type of adaptation is the hard covering?

[1 mark]

Tick (✓) **one** box.

Behavioural

☐

Functional

☐

Structural

☐

Turn over ►



0 2 . 1

Cells contain genes.

Which chemical are genes made from?

[1 mark]Tick (✓) **one** box.

Carbohydrate

☐

DNA

☐

Protein

☐**0 2 . 2**

Genes are passed on to offspring in reproduction.

Which statement is correct for **sexual** reproduction?**[1 mark]**Tick (✓) **one** box.

Gametes are involved

☐

Offspring are clones

☐

Only one parent is needed

☐

Reproduction requires animals to communicate.

Animals communicate with each other in different ways.

0 2 . 3 What is a type of signal animals use to communicate?

[1 mark]

Tick (✓) **one** box.

Antibodies

☐

Sounds

☐

Vaccines

☐

Question 2 continues on the next page

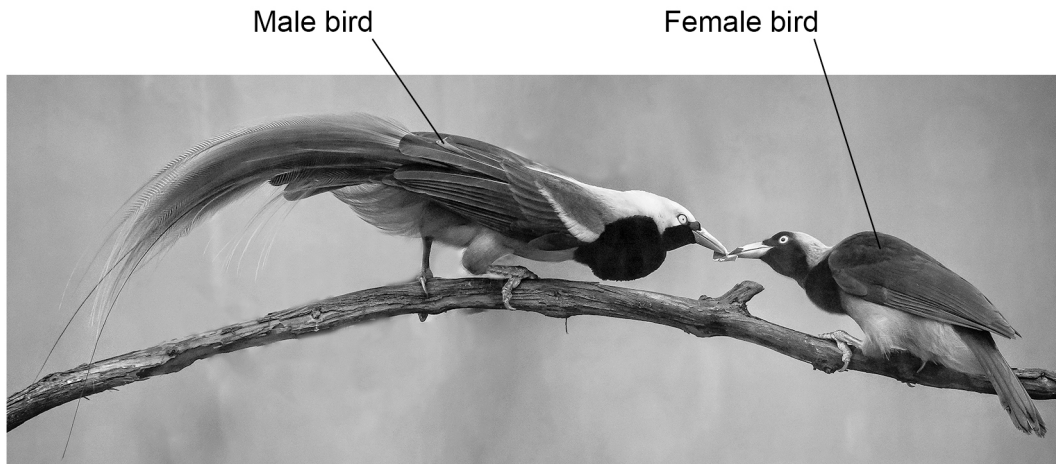
Turn over ►



Signals can also be visual.

Figure 3 shows two birds of the same species.

Figure 3



0 2 . 4

Describe **two** differences between the appearance of the male bird and the female bird.

[2 marks]

1 _____

2 _____



0 2 . 5 Animals display different types of behaviour.

Draw **one** line from each description to the type of behaviour that it shows.

[2 marks]

Description	Type of behaviour
	Classic conditioning
A dog releases saliva when a bell rings	Habituation
	Imprinting
A spider spins a web	Innate
	Operant conditioning

7

Turn over for the next question

Turn over ►



0 3 . 1 Living organisms are made of cells.

Each part of a cell has a different function.

Draw **one** line from each cell part to the description for that part.

[3 marks]

Cell part	Description
	Loops of DNA containing genes
Mitochondria	Structure to control what goes into and out of cell
Plasmids	Structure where sugar is made from carbon dioxide and water
Ribosomes	Structure where energy is released
	Site of protein production

0 3 . 2 Gases move into cells by diffusion.

What is diffusion?

[2 marks]



A student investigated the effect of different concentrations of sugar solution on plant cells.

This is the method used.

1. Label three tubes **A**, **B** and **C**.
2. Add a different concentration of sugar solution to each tube.
3. Place a thin layer of cells from the same plant into each tube.
4. Leave the tubes in the same room for 24 hours.
5. Remove the cells and view them through a microscope.

0 3 . 3 What was the independent variable in this investigation?

[1 mark]

Tick (✓) **one** box.

Concentration of the sugar solution

☐

Temperature of the room

☐

Time cells left in the sugar solution

☐

0 3 . 4 During an investigation it is important to recognise that errors can occur.

Complete the sentence.

[1 mark]

Choose the answer from the box.

random errors

systematic errors

zero errors

Repeating the investigation and calculating a mean reduces

the effect of _____.

Turn over ►



Figure 4 shows a cell before being placed in the sugar solution.

Figure 4

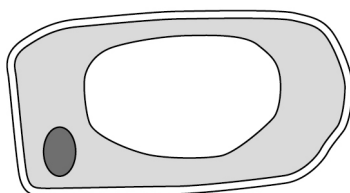


Table 1 shows the results.

Table 1

Tube	Appearance of one cell
A	
B	
C	

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0 3 . 5 Explain the appearance of the cell in tube **C**.

[4 marks]

0 3 . 6 An **animal** cell was placed in the sugar solution in tube **A**.

The result for the animal cell was different from the result for the plant cell.

Explain why the result for the animal cell was different.

[2 marks]

Result _____

Explanation _____

13

Turn over for the next question

Turn over ►

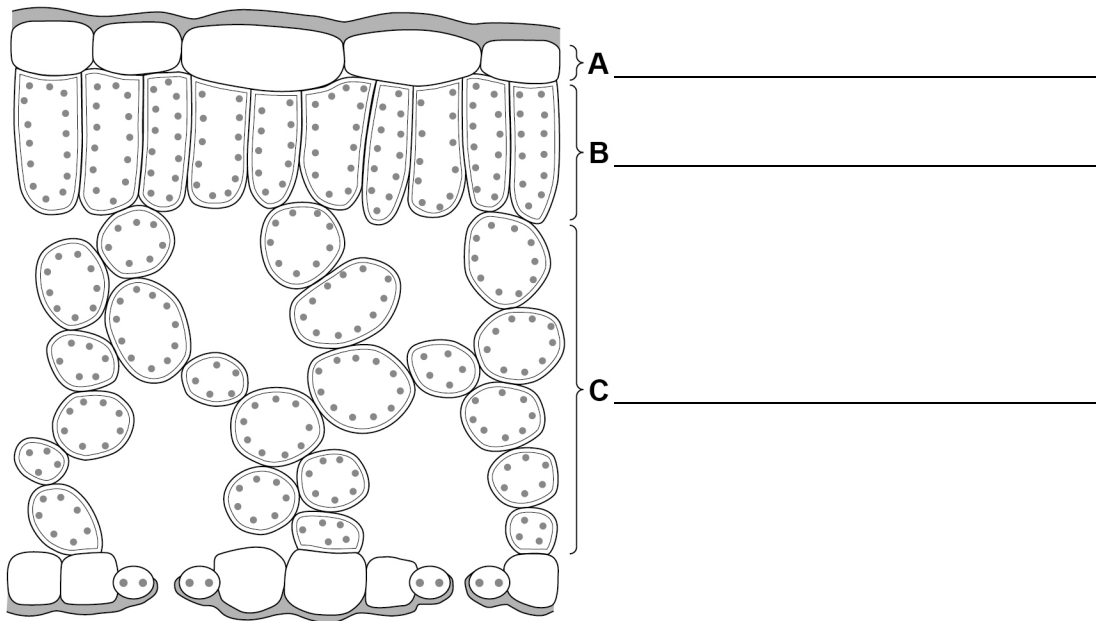


0 4

A leaf is a plant organ.

0 4

. 1

Figure 5 shows a section through a leaf.**Figure 5**Label layers **A**, **B** and **C** on **Figure 5**.**[3 marks]**

Choose the answers from the box.

Epidermal tissue**Palisade mesophyll****Spongy mesophyll****Stomata****Xylem**

0	4	.	2
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Some substances in plants are transported by translocation.

Describe the process of translocation.

[4 marks]

Question 4 continues on the next page

Turn over ►



The baobab tree grows in Madagascar, Africa.

Temperatures in Madagascar are consistently high throughout the year.

Table 2 shows the mean monthly rainfall in Madagascar over 10 years.

Table 2

Month	Mean rainfall in millimetres
January	240
February	200
March	90
April	15
May	11
June	2
July	2
August	2
September	4
October	12
November	20
December	165

0 4 . 3 Describe how the mean rainfall for one month would have been calculated.

[2 marks]



0 4 . 4

Leaves are **only** on the baobab tree from December to February.

Explain why having leaves only from December to February helps the baobab tree to survive.

Use information from **Table 2** in your answer.

[4 marks]

13

Turn over for the next question

Turn over ►



0 5

Pathogens are microorganisms that cause infectious diseases.

A virus is a type of pathogen.

People can be vaccinated against diseases caused by viruses.

0 5**1**

What does a vaccine contain that stimulates the production of antibodies?

[1 mark]

0 5**2**

The MMR vaccine is used to protect children against measles.

Name the **two** other diseases the MMR vaccine protects against.

Do **not** refer to measles.

[2 marks]

1

2



A vaccine is given to protect against the influenza virus.

The vaccine stimulates antibody production against three different types of the influenza virus, **A**, **B** and **C**.

Scientists investigated whether the time of day a vaccination was given affected the concentration of antibodies produced.

This is the method used.

1. Select 274 adults above 65 years of age.
2. Measure the mean concentration of antibodies for each type of influenza. These are the baseline antibody concentrations.
3. Divide the adults into two groups:

Group 1 — receive the vaccination in the morning.

Group 2 — receive the vaccination in the afternoon.
4. Measure the mean concentration of antibodies for each group 3 months after the vaccination.

0 5 . 3

Give **two** control variables that should have been used in the trial.

Do **not** refer to age.

[2 marks]

1 _____

2 _____

Question 5 continues on the next page

Turn over ►



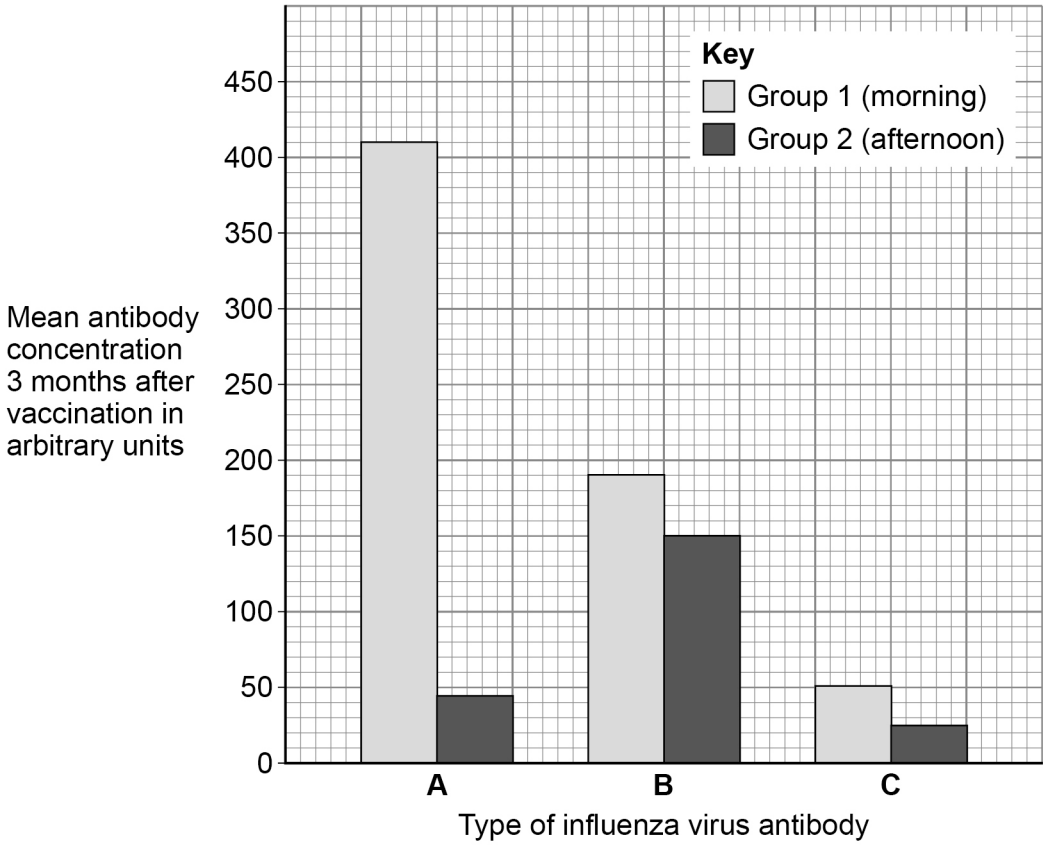
Table 3 shows the mean baseline antibody concentrations in arbitrary units.

Table 3

Type of influenza virus antibody	Mean baseline antibody concentration in arbitrary units
A	35
B	45
C	8

Figure 6 shows the results of the trial.

Figure 6



Scientists concluded that having the vaccine in the morning was more effective compared to having the vaccine in the afternoon.

[5 marks]

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0 5 . 5 A different trial was carried out.

The trial compared people with and without an immune disorder.

People with the immune disorder do **not** produce many white blood cells.

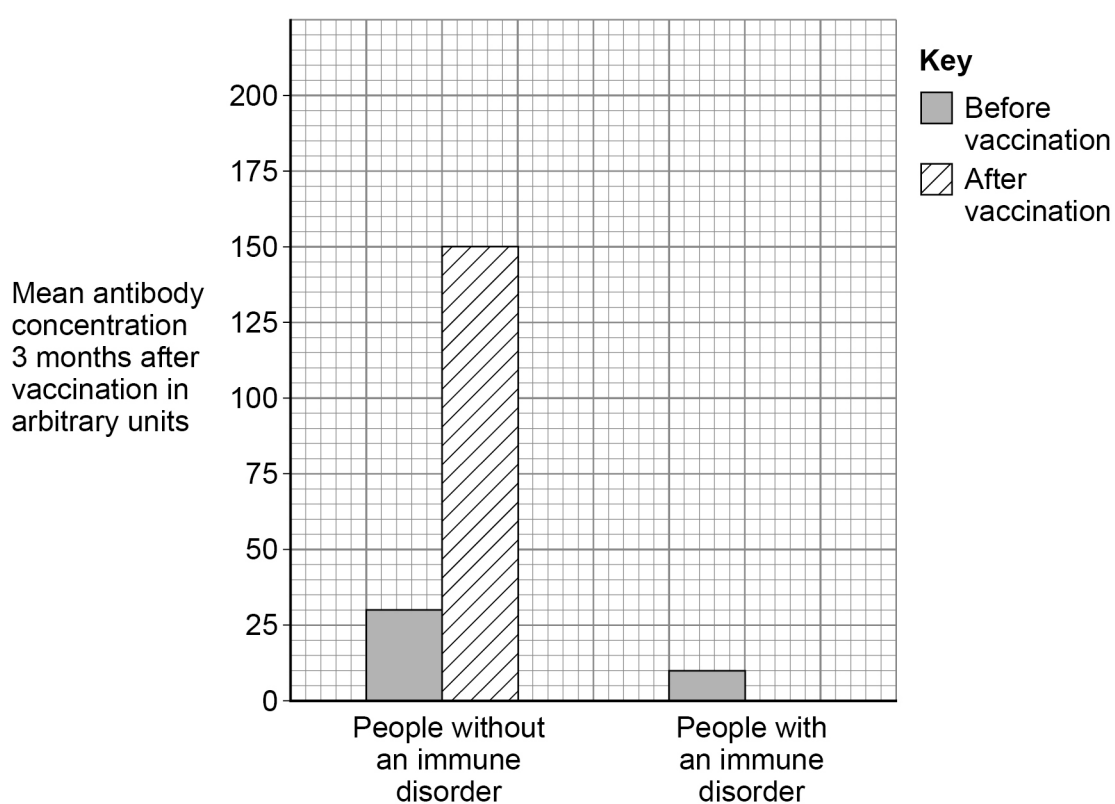
Complete **Figure 7**.

You should:

- draw a bar to show the result you would expect after vaccination from people with an immune disorder.

[1 mark]

Figure 7



0 5 . 6 Why is it difficult to develop drugs to destroy the influenza virus?

[1 mark]



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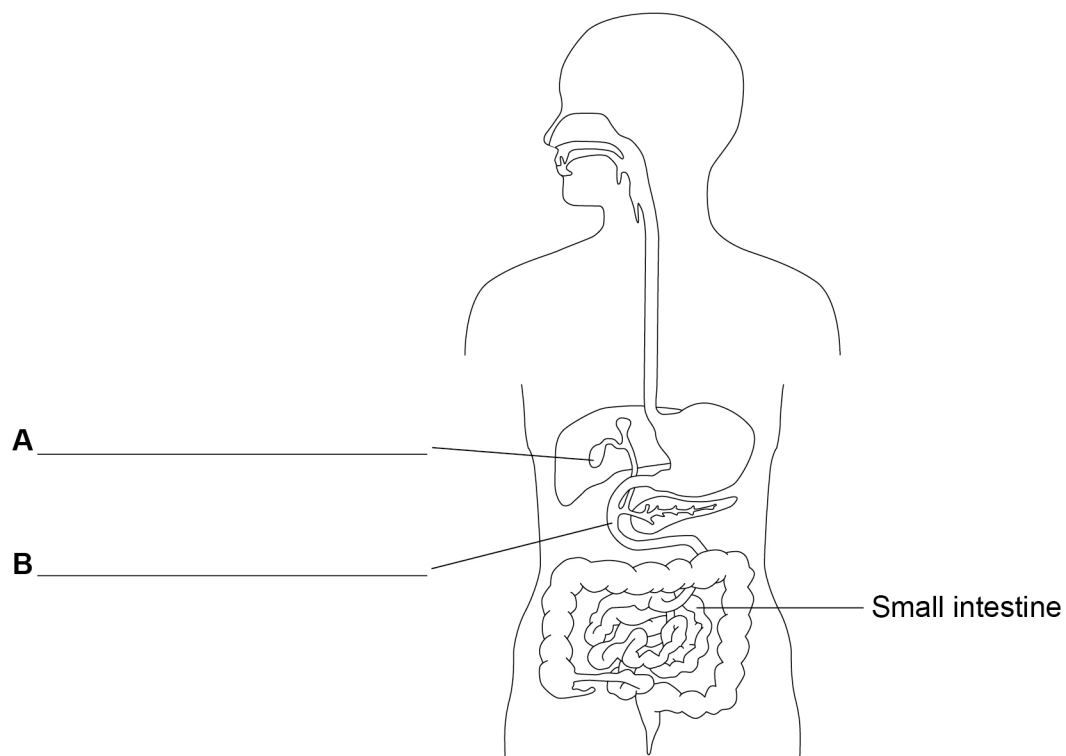


0 6

This question is about the digestive system.

Figure 8 shows organs in the digestive system.

Figure 8

**0 6****. 1**

Label **A** and **B** on **Figure 8**.

[2 marks]



[6 marks]

[illegible]

Turn over ►



Table 4 shows information about the diets of two people.

Table 4

Nutrient	Mass of nutrient consumed in milligrams per day	
	Person 1	Person 2
Polyphenols	1050.0	420.0
Vitamin A	0.7	0.2
Sugar	24 000.0	65 000.0
Sodium	3 300.0	3 400.0
Oils	500.0	15 700.0

0 6 . 3 Calculate how much more polyphenols are in the diet of **Person 1** compared to **Person 2**.

Give your answer as a percentage.

[3 marks]

Percentage = _____ %



[5 marks]

[illegible]

16

Turn over for the next question

Turn over ►



0 7

The nervous system responds to information from receptors.

0 7 . 1

Name **two** sense organs which have receptors sensitive to chemicals.

[2 marks]

1 _____

2 _____

0 7 . 2

Muscles respond to information from receptors in a reflex action.

Describe the pathway of a nerve impulse in a pain-withdrawal reflex.

[4 marks]



0 7 . 3 One inherited disorder affects muscles.

The ratio of people with the disorder is 7 males : 4 females.

1 in 50 000 people is expected to have the disorder.

A population has 59 500 000 people.

Calculate how many **females** would be expected to have the disorder in the population.

[4 marks]

Number of females = _____

Question 7 continues on the next page

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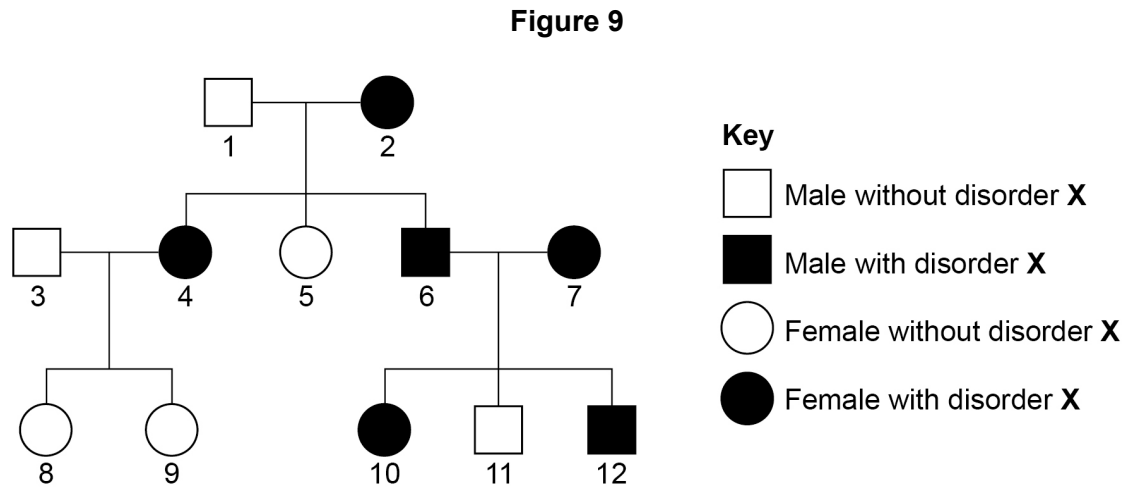


Disorder **X** is a different inherited condition.

Disorder **X** causes people to have a dangerous reaction to certain drugs.

The reaction is due to differences in their muscle tissue.

Figure 9 shows the inheritance of disorder **X** in one family.



0 7 . 4 Give **one** piece of evidence to show that disorder **X** is caused by a dominant allele.

Use **Figure 9**.

[1 mark]

0 7 . 5

Persons **3** and **4** in **Figure 9** are having a third child.Determine the probability that the child will have disorder **X**.

You should:

- complete the genetic diagram
- identify the phenotype of each offspring genotype
- use the symbols:

D = dominant allele for disorder **X****d** = recessive allele for not having disorder **X****[5 marks]**

		Female	
Male			

Probability of the child having disorder **X** = _____**Question 7 continues on the next page****Turn over ►**

0 7 . 6

A drug stimulates the muscles in people who have disorder **X**.

The drug causes the body temperature of people with disorder **X** to rise to dangerous levels.

Explain why the body temperature of a person with disorder **X** increases when they are given the drug.

[3 marks]

0 7 . 7

Sustained muscle stimulation can cause muscle cell death.

When cells die they release excess protein into the blood.

Proteins are too large to be filtered by the kidneys.

Suggest how the death of muscle cells might cause kidney failure.

[1 mark]

20

END OF QUESTIONS



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